$$e_k(\zeta^j A) = \zeta^{jk} e_k(A)$$

	k = 1	k = 2	k = 3
j = 0	1	1	1
j = 1	ζ	ζ^2	1
j = 2	ζ^2	ζ	1

Rotating factors only. A zero coefficient remains zero.